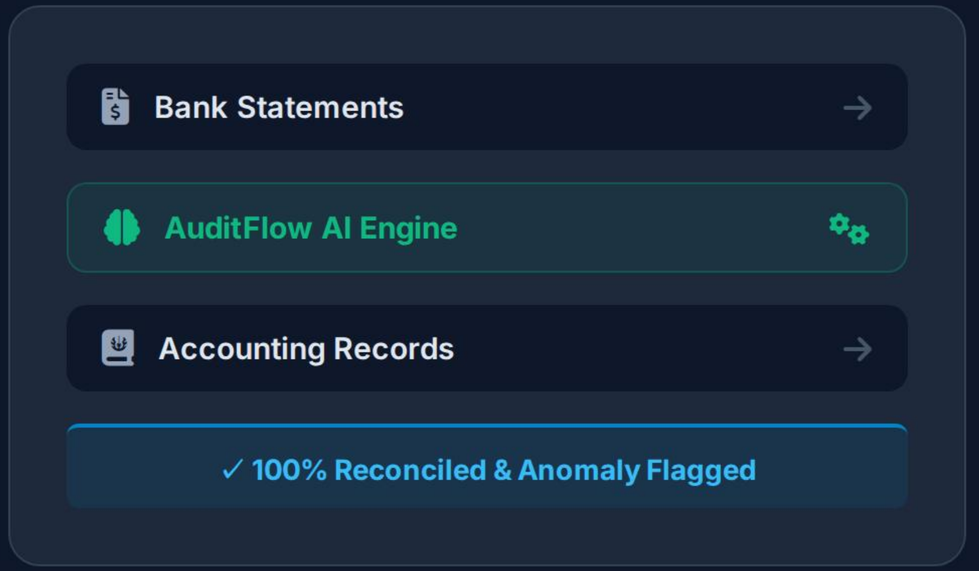
AI-POWERED FINANCIAL RECONCILIATION & INTELLIGENT AUDIT

HACKATHON PRESENTATION

AUDITFLOW AI

Reconcile. Investigate. Resolve.

Automated bank statement reconciliation & anomaly investigation agent powered by deterministic signals and local LLMs.



TEAM NAME

[Insert Team Name]

TEAM MEMBERS

Member 1 • Member 2 • Member 3 • Member 4

CHALLENGE & OPPORTUNITY

Problem Statement & Project Overview

The Reconciliation Bottleneck

Organizations maintain fragmented records across banks, ERPs, and spreadsheets resulting in frequent discrepancies:

- **Transaction Gaps:** Unrecorded entries and missing fees
- **Data Mismatches:** Duplicate entries, amount & date variances
- **Unclear Descriptions:** Vague wire references and bank charges
- **Fraud Risk:** Undetected high-value or suspicious anomalies

Manual Flow: Bank PDF → Excel → Manual Match → Slow, error-prone & non-scalable.

The AuditFlow AI Solution

An end-to-end intelligent agent that automates reconciliation and audit investigations:

1. Automated Comparison
Multi-signal matching across amounts, dates, and references.

2. Anomaly Classification
Flags duplicates, timing offsets, and suspicious line items.

3. AI Investigation Agent
Generates step-by-step evidence and clear resolution advice.

Result: 90%+ reduction in audit cycle times with full explainability.

ARCHITECTURE & TOOLS

Technology Stack & Rationale

Frontend & Dashboard

Next.js & React: Responsive UI

TypeScript: Type safety

Tailwind CSS: Clean styling

Recharts: Audit visualizers

Lucide Icons: UI elements

Backend & Processing

Python & FastAPI: Core REST APIs

Pandas & NumPy: Data engine

PyMuPDF: Bank PDF parsing

RapidFuzz: String scoring

SQLite & SQLAlchemy: Data store

AI & Testing

Ollama: Local LLM host

Local LLMs: Private reasoning

Pytest: Comprehensive testing

FastAPI TestClient: API verification

Git & VS Code: CI & Dev workflow

Why Local AI (Ollama)?

Zero API Costs & Total Privacy: Sensitive financial data stays strictly on-premise. Deterministic rule-based algorithms handle financial matching, while Ollama handles complex investigation reasoning.

PIPELINE & LOGIC

Detailed Project Workflow

STEP 1 & 2

Ingestion & Parsing

Extract PDF/CSV statement data: Date, Description, Amount, Type & UTR reference.

STEP 3

Normalization

Standardize date formats, clean vendor strings, and normalize currency values.

STEP 4

Multi-Signal Match

Correlate records using combined signals: Amount, Date, Description, and UTR.

Match Confidence Scoring Model: **Amount:** 40% **Date Proximity:** 20% **Description:** 20% **Reference/UTR:** 20%

STEP 6

Anomaly Detection

Identify missing records, duplicates, timing shifts, and high-value risks.

STEP 7

AI Investigation Agent

Executes multi-step checks: Ref → Amount → Range → Similarity → Classification.

STEP 8 & 9

Explainable Output

Delivers structured evidence, confidence rating, and dashboard audit report.

CORE CAPABILITIES

Key Features & AI Intelligence

Automated Matching

Replaces manual spreadsheets with automated cross-system bank comparison.

Confidence Scoring

Evaluates financial and textual signals to assign a deterministic match rating.

Anomaly Detection

Automatically categorizes duplicates, missing entries, and timing shifts.

AI Investigation Agent

Performs multi-stage reasoning to analyze and resolve complex edge cases.

Explainable Results

Clearly separates objective FACTS from AI guidance and recommendations.

Privacy-First Local AI

Uses Ollama LLMs locally to safeguard confidential banking data.

VALUE & ROADMAP

Business Impact & Future Scope

Expected Business Impact

- **80%+ Time Reduction:** Minimizes manual reconciliation overhead.
- **Match Consistency:** Standardizes scoring rules across statements.
- **Traceable Audits:** Provides step-by-step reasoning logs.
- **Data Security:** Protects sensitive records via local AI processing.

Centralized Dashboard Insights for CFOs & Auditors

Future Roadmap

Direct ERP Integration
Native connectors for Tally, Zoho Books, and SAP.

Advanced OCR Support
Parse low-quality scanned paper receipts & bank slips.

Predictive Fraud Analytics
Detect suspicious velocity spikes and recurring risk profiles.

Enterprise-Ready • Role-Based Access Control (RBAC)

THANK YOU

AUDITFLOW AI

Reconcile. Investigate. Resolve.
Automated Bank Statement Reconciliation & Anomaly Investigation Agent

Questions & Live Demonstration

We would be happy to demonstrate how AuditFlow AI detects and explains financial discrepancies.

Team: **[Insert Team Name]** • Members: **Member 1 | Member 2 | Member 3**